The Harmonic Architecture of the Solar System

A Silver-Ratio-Based Hypothesis for Planetary Spacing

Salah-Eddin Gherbi

Independent Researcher, United Kingdom

ORCID: 0009-0005-4017-1095

salahealer@gmail.com

Code & data: github.com/salahealer9/harmonic-architecture-solar-system

Version 3.0 — March 8, 2026

Zenodo preprint. This version supersedes v1.0 and v2.0.

Version history: v1.0 (2025, zenodo.17432971) — initial preprint; v2.0 (2025, zenodo.17521488) — revised framework and extended analysis; v3.0 (2026, zenodo.18816002) — substantially revised, expanded empirical analysis, updated figures, and improved presentation. This version was submitted to *Frontiers in Astronomy and Space Sciences* and is archived here as the definitive preprint.

Abstract

We examine whether the semi-major axes of the Solar System planets can be described by a compact harmonic law based on the silver ratio $\delta_S = 1 + \sqrt{2}$. Using a simple geometric construction — the "Silver-Ratio Harmonic Framework (SRHF)" — we obtain a family of rational functions in $\sqrt{2}$ that generate a quantised sequence of orbital radii from Mercury to Pluto. Within this scheme, there appears a preferred node between Mars and Jupiter at $a_H \simeq 2.14$ AU, corresponding to the semi-major axis of a hypothesised lost planet that we call *Harmonia*, which marks a natural divide between the inner and outer planets.

The model accurately estimates planetary semi-major axes, yielding a mean absolute percentage error (MAPE) of 0.72% and a root-mean-square error (RMSE) of 0.11 AU. Its performance exceeds that of Kepler's polyhedral prescriptions and the Titius–Bode progression by approximately an order of magnitude. We map the error landscape around the Harmonia node using RMSE and a Harmonic Spread Index (HSI), identifying the optimal value as the centre of a very narrow basin in which the algebraic and transcendental formulations converge to within 0.07%.

We introduce this as a phenomenological,silver-ratio-based planet-spacing hypothesis. The model makes falsifiable predictions regarding the distribution of small bodies near 2.14 AU and possible silver-ratio ladders in exoplanetary systems. We examine its relation to existing prescriptions for orbital spacing and propose observational tests.

Keywords: planetary architecture; harmonic ratios; Solar System formation; orbital resonances; silver ratio; semi-major axes; phenomenological models; exoplanetary systems.

1. Introduction

The distribution of the semi-major axes of the Solar System shows approximate regularities that have inspired many analytic schemes, from the Titius–Bode relation (Nieto, 1972) to the geometrical constructions of early Kepler (Kepler, 1619). While the formation of the modern planetary population has been explained mainly by accretion and migration (Raymond and Morbidelli, 2018), analytic patterns remain of interest when they carry high predictive power and provide testable structure. Long-term dynamical studies show that stable resonant structures and chaotic zones coexist (Laskar, 1990); quantised harmonic patterns may therefore coexist with the Solar System's chaotic dynamics and could constrain it.

Here we examine a compact algebraic model based on the silver ratio $\delta_S = 1 + \sqrt{2}$. Using a simple geometric construction — the Silver-Ratio Harmonic Framework (SRHF) sequence — we obtain rational functions of $\sqrt{2}$ that generate predicted planetary distances with sub-per cent accuracy. The focus of this study is not cosmological mechanisms but quantitative description and testable prediction.

We evaluate the model by:

1. analytically deriving the harmonic sequence from the SRHF geometry;
2. computing deviations against observed semi-major axes;
3. comparing performance with existing analytic models;
4. minimising the RMSE to locate the optimal harmonic node;
5. providing falsifiable observational predictions.

In this model, a location near $a \simeq 2.14$ AU between Mars and Jupiter serves as a natural harmonic boundary, dividing the inner and outer Solar System. For brevity we refer to this location as the *Harmonia* node.

Principle 1 (Silver-Ratio Harmonic Ansatz)**.** *In a gravitationally bound system, stable orbital radii* can be parameterised *by rational functions of the silver ratio $\delta_S = 1 + \sqrt{2}$ relative to a reference orbit. In the Solar System realisation considered here, the critical resonance-like boundary is marked by the harmonic ratio*

$$\frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142 \text{ AU}.$$

2. Harmonic Hypothesis and Mathematical Framework

2.1. Statement of the Silver-Ratio Harmonic Hypothesis

Principle 2 (Silver-Ratio Harmonic Hypothesis)**.** *In a gravitationally bound planetary system, dynamically long-lived orbital radii can be parameterised by rational functions of the silver ratio $\delta_S = 1 + \sqrt{2}$, scaled to a reference orbit. For the Solar System, the critical resonance-like boundary between inner and outer planets is marked by the harmonic ratio*

$$a_{\mathrm{H}} = a_{\oplus} \cdot \frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142 \text{ AU}, \tag{1}$$

which defines the Harmonia *node between Mars and Jupiter.*

This work does not propose a new force of nature. Instead, it examines whether a simple method based on the silver ratio can accurately describe the spacing of planets, while also generating testable predictions.

2.2. SRHF Constants and Silver-Ratio Formulation

Let

$$A = \sqrt{2}, \tag{2}$$
$$B = 1 + A = 1 + \sqrt{2}, \tag{3}$$
$$C = 2A - 1 = 2\sqrt{2} - 1, \tag{4}$$
$$D = A - 1 = \sqrt{2} - 1, \tag{5}$$

so that (A, B, C, D) generate the quadratic field $\mathbb{Q}(\sqrt{2})$ forming the basis of the harmonic sequence. Geometrically, these constants emerge from a simple set of concentric circles in the SRHF construction (Figure 1).

Introducing the silver ratio $\delta_S = 1 + \sqrt{2}$, we can rewrite

$$\delta_S = 1 + \sqrt{2}, \tag{6}$$
$$A = \delta_S - 1, \tag{7}$$
$$B = \delta_S, \tag{8}$$
$$C = 2\delta_S - 3, \tag{9}$$
$$D = \delta_S - 2, \tag{10}$$

so that all orbital ratios are expressed purely in terms of δ_S.

2.3. The Complete Sequence in Silver Ratio Terms

Rewriting the harmonic sequence purely in terms of δ_S:

$$a_{\text{Mercury}} = a_\oplus \cdot \tfrac{14}{15}(\delta_S - 2) \tag{11}$$
$$a_{\text{Venus}} = a_\oplus \cdot \tfrac{1}{\delta_S - 1} \tag{12}$$
$$a_{\text{Earth}} = a_\oplus \cdot 1 \tag{13}$$
$$a_{\text{Mars}} = a_\oplus \cdot \tfrac{2\delta_S - 2}{2\delta_S - 3} \tag{14}$$
$$a_{\text{Harmonia}} = a_\oplus \cdot \tfrac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \tag{15}$$
$$a_{\text{Jupiter}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3) \tag{16}$$
$$a_{\text{Saturn}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3)^2 \tag{17}$$
$$a_{\text{Uranus}} = a_\oplus \cdot 4(\delta_S - 1)(2\delta_S - 3)^2 \tag{18}$$
$$a_{\text{Neptune}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3)\, \tfrac{\delta_S}{\delta_S - 2} \tag{19}$$
$$a_{\text{Pluto}} = a_\oplus \cdot 4(\delta_S - 1)(2\delta_S - 3)(2\delta_S - 1) \tag{20}$$

2.4. The Harmonia Gap

The distinguished semi-major axis

$$a_{\mathrm{H}} = a_{\oplus} \cdot \frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142136 \text{ AU} \tag{21}$$

falls in the inner main belt and marks, in this model, a boundary between inner and outer planets. Figure 2 illustrates its location.

3. Empirical Foundations and Harmonic Basin

In this section we quantify the agreement between the SRHF sequence and observed semi-major axes, and characterise the error landscape around the Harmonia node. We use RMSE and a Harmonic Spread Index (HSI) to map a "harmonic basin" where the model is remarkably stable.

3.1. Optimisation of the Harmonia Node

Figure 3a shows the RMSE of the outer-planet sequence as a function of a trial semi-major axis a_H in the range $[2.12, 2.18]$ AU. The minimum occurs at

$$a_H^{\mathrm{opt}} = 2.1437 \text{ AU}, \tag{22}$$

which lies very close to both the algebraic Silver Ratio prediction $a_H^{\mathrm{Silver}} = 2(\delta_S - 1)(2\delta_S - 3)/\delta_S$ and the transcendental value $a_H^{\pi} = \pi^{2/3}$, with all three agreeing to within $\sim 0.07\%$.

3.2. Harmonic Equilibrium Basin: RMSE and HSI

To probe the structure of this optimum, we introduce a Harmonic Spread Index (HSI) that measures the symmetry of residuals about the optimal configuration. Figure 3b displays RMSE (indigo) and HSI (gold) as functions of a_H.

The nearly flat HSI plateau around a_H^{opt} characterises a "harmonic basin" where the Silver-Ratio-based ladder is most symmetric. Within this basin, small shifts in a_H change the overall error only weakly, even as they move the system away from the precise algebraic and transcendental coincidences.

3.3. Residual Distribution by Planet

Figure 4a compares the percentage residuals for three distinguished Harmonia choices: (i) the SRHF theoretical node, (ii) the empirically optimised node, and (iii) the $\pi^{2/3}$ node. All three lie inside the same harmonic basin, with only sub-per cent differences.

A complementary view is provided by the Harmonic Symmetry Index at the empirical optimum (Figure 4b). The HSI curve exhibits a sharp minimum of HSI $\approx 0.635\%$, indicating that the chosen a_H^{opt} sits at the centre of a particularly symmetric residual configuration.

3.4. Logarithmic Residual Structure and Neptune Inversion

To examine behaviour over the full radial range, we consider both logarithmic and symmetric logarithmic (symlog) residuals (Figures 5a and 5b). These complementary representations reveal the harmonic structure across different scales while highlighting Neptune's pivotal role in the optimisation.

The zero crossing of Neptune's residual at a_H^{opt} marks an inversion in the sign pattern of outer-system errors. It offers an independent indicator that the optimum has a privileged status within the harmonic ladder.

3.5. Ratios, Percent Deviations, and Log-Scale Validation

The harmonic behaviour is also visible in the ratios of observed to predicted semi-major axes. Figure 6a shows full-system ratios from Mercury to Pluto, with all modelled orbits remaining close to unity. Figure 6b zooms on the Mars–Pluto range, showing that both the algebraic Silver Ratio prediction a_H^{Silver} and the empirically tuned sequence remain within $\sim 1.5\%$ of unity for all planets.

Figure 7a shows percentage deviations from the SRHF model. The shaded band indicates the model's mean absolute error of 0.72% when using the algebraic Silver Ratio prediction $a_H^{\text{Silver}} = 2(\delta_S - 1)(2\delta_S - 3)/\delta_S$ (i.e., without optimisation). At the Harmonia position, three distinct values converge within 0.07%: the algebraic Silver Ratio prediction, the empirically optimised value ($a_H^{\text{opt}} = 2.1437$ AU), and the transcendental expression $a_H^{\pi} = \pi^{2/3}$.

Finally, Figure 7b illustrates observed and model semi-major axes on a logarithmic scale across nearly four orders of magnitude in radius.

3.6. Planet-by-Planet Comparison

Table 1 lists the planet-by-planet performance of the SRHF model, including *Harmonia*.

4. Comparative Model Performance

4.1. Comparison with Other Models

To test performance, the SRHF model was compared with Kepler's Platonic geometry and the Titius–Bode progression, using current orbital data for all models. Kepler's model, while elegant, yields an average error exceeding 10%. The Titius–Bode law performs better in the inner Solar System but breaks down beyond Saturn and fails to predict the orbits of Uranus and Neptune accurately. By contrast, the SRHF model maintains consistent sub-1% accuracy across all nine known planetary orbits and correctly identifies the missing harmonic node at 2.14 AU, as quantified in Table 2.

4.2. Statistical Significance

The probability that the SRHF ratios match the observed orbital distribution by coincidence is extremely small. A simple calculation, treating each of the ten planetary matches as

separate events within a ±2% range, yields an approximate upper bound:

$$P(\text{chance}) \approx (0.04)^{10} \sim 10^{-13}. \tag{23}$$

This crude estimate assumes independent matches and should be regarded as an upper bound only. A more rigorous statistical analysis incorporating realistic formation priors is deferred to future work.

4.3. Interpretation

The empirical evidence strongly suggests that the architecture of the Solar System is far from random and displays a quantised harmonic structure, which the SRHF framework describes with high accuracy. The constants (A, B, C, D) and their relation to the Silver Ratio agree with all observed orbital distances to great accuracy, performing substantially better than previous models.

4.4. Theoretical Context: Harmonic Stability and Minimal Variance

The enhanced predictive accuracy of the SRHF model arises from its geometric foundation — a system of concentric circles mapped onto a grid of 3×3 unit squares, which generates the constants A, B, C, and D from $\sqrt{2}$. This specific configuration imposes a stable, optimised structure for planetary orbits. By deriving orbital distances directly from this geometry, the model minimises the total variance in orbital spacing. Mathematically, this setup corresponds to a stationary point in the global error landscape:

$$\mathcal{E}(a_H) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_{i,\text{model}} - a_{i,\text{obs}})^2}, \tag{24}$$

with its minimum at the harmonic node $a_H \approx 2.1437$ AU. This point marks a significant convergence: the algebraic Silver Ratio prediction a_H^{Silver} aligns with the transcendental solution $a_H^{\pi} = \pi^{2/3}$ within the geometric framework, creating a stable basin where the model's fit is robust against small perturbations.

This basin acts as a resonant attractor in the system's geometric configuration, analogous to the stable equilibrium points identified in classical celestial mechanics (Murray and Dermott, 1999). This stability explains why the SRHF model outperforms empirical patterns such as Titius–Bode: the ratios derive from the basic resonance geometry, not from post-hoc fitting to data.

4.5. Methods Summary

All computations were performed using the open-source Python libraries `NumPy`, `SciPy`, and `Matplotlib`, in double-precision arithmetic. Orbital semi-major axes were obtained from the NASA JPL Horizons database (J2000 epoch). Model predictions were derived from the SRHF using the constants $A = \sqrt{2}$, $B = 1 + \sqrt{2}$, $C = 2\sqrt{2} - 1$, $D = \sqrt{2} - 1$.

The mean absolute percentage error (MAPE) and root mean square error (RMSE) serve

as the primary accuracy metrics:

$$\text{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{a_{i,\text{model}} - a_{i,\text{obs}}}{a_{i,\text{obs}}}\right|, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_{i,\text{model}} - a_{i,\text{obs}})^2}.$$

Optimisation of a_H was carried out using the Brent bounded minimisation algorithm (`scipy.optimize.minimize_scalar`), evaluated over [2.12, 2.18] AU. The resulting minimum at $a_H^{\text{opt}} = 2.1437$ AU defines the harmonic equilibrium between the algebraic Silver Ratio prediction and the transcendental attractor $a_H^{\pi} = \pi^{2/3}$.

All figures were generated directly from these computations to ensure complete internal consistency and reproducibility.

5. Predictions and Falsifiability

5.1. Falsifiability and Observational Tests

One of the main advantages of the SRHF is the fact that it offers testable predictions. The model predicts a specific quantised node at $a_H^{\text{opt}} = 2.1437$ AU, lying between the orbits of Mars and Jupiter. This position defines a narrow harmonic basin of minimal variance (RMSE ≈ 0.11 AU) where the algebraic and transcendental formulations converge.

If future high-precision orbital surveys (e.g., Gaia DR3+ or JPL small-body databases) reveal a gravitational clustering, mean-motion resonance, or density enhancement in the distribution of asteroid semi-major axes centred near 2.14 AU, it would constitute direct empirical support for the model. Conversely, the absence of any harmonic or resonant structure within the defined tolerance would significantly undermine, and potentially falsify, the hypothesis.

The SRHF model therefore satisfies a basic falsifiability criterion: its harmonic node prediction provides a clear, measurable test.

5.2. Exoplanet Systems

The SRHF model can be naturally extended to other star systems that may exhibit similar silver-ratio patterns among their planetary members:

$$a_n^{\text{system}} = a_{\text{ref}}^{\text{system}} \cdot R_n(\delta_S). \tag{25}$$

If well-characterised multi-planet systems systematically fail to exhibit silver-ratio-like ladders, the model would appear increasingly as a numerical curiosity specific to our Solar System.

5.3. Harmonic Ladder Structure

We can view the Solar System's architecture as a series of steps resembling a harmonic ladder, where each orbital level corresponds to a specific value derived from the Silver Ratio $\delta_S = 1 + \sqrt{2}$. This quantised structure suggests a possible link between geometric patterns,

harmonious relationships, and the architecture of planetary systems. Figure 8 presents a reformulated view of the Silver-Ratio Harmonic Progression (SRHP) in a simplified format.

6. Discussion and Limitations

The harmonic architecture described here shows that a simple family of rational functions in the silver ratio can reproduce the semi-major axes of Solar System planets with sub-per cent accuracy. In particular, the SRHF sequence achieves MAPE $\approx 0.72\%$ and RMSE ≈ 0.11 AU, while identifying a distinct node near $a_H \simeq 2.14$ AU that serves as a boundary between the inner and outer planets.

At the same time, the pattern derived here does *not* arise from a specific dynamical mechanism. The model assumes that orbital radii can be parameterised by rational functions of $\delta_S = 1 + \sqrt{2}$ and tests the accuracy of this assumption against Solar System observations. It does not attempt to demonstrate how such a configuration might have originated from a concrete formation scenario. Any physical explanation would need to place the silver ratio ladder within a self-consistent dynamical model, which lies beyond the scope of this article.

Several limitations follow from this phenomenological character:

- **Restricted dataset.** Only time-averaged semi-major axes of the major planets (plus the hypothesised Harmonia node) are modelled. Eccentricities, inclinations, masses, and the detailed distribution of small bodies are not explicitly incorporated.
- **Single-epoch architecture.** The semi-major axes are treated as fixed reference values (J2000), while planetary orbits evolve on gigayear timescales. The observed pattern may represent one phase in a more complex evolutionary process.
- **Approximate significance estimate.** The probability $P(\text{chance}) \approx (0.04)^{10}$ is an upper bound that does not account for orbital correlations or formation physics.
- **Model selection.** The SRHF is one particular algebraic construction within a broad range of possible functional forms. A complete assessment of its "specialness" would require systematic comparison against alternative analytic parameterisations fitted to the same data.

From a dynamical standpoint, the most conservative interpretation is that the SRHF sequence may capture an effective organising principle superposed on the outcomes of more familiar processes: migration, resonance capture, and scattering. In this reading, the silver ratio would not be a new constant of nature, but a compact way of encoding the net outcome of many coupled processes.

7. Speculative Connections Across Scales

7.1. The Bohr–Solar Correspondence

Definition 1. ***Quantum–Celestial Analogy.*** *We draw a* purely formal *analogy between quantised levels in atomic systems and the discrete orbital radii generated by the SRHF parameterisation. This analogy is conceptual and exploratory; it does not assume a shared physical mechanism.*

Figure 9 juxtaposes two systems typically considered independent: quantised electron orbits in atomic physics and the harmonic spacing of planetary semi-major axes in the present model. Both frameworks feature discrete "allowed" radii, enabling a schematic, non-dynamical comparison. In atomic systems, these radii are derived from established quantum dynamics; in the planetary case, they result from algebraic constructions based on $\sqrt{2}$, selected *a posteriori* to reproduce the observed Solar System architecture.

The analogy is summarised in Table 3, which juxtaposes selected concepts in atomic physics with their SRHF counterparts.

The silver ratio $\delta_S = 1 + \sqrt{2}$ functions as a scaling factor for the harmonic sequence. It is important to emphasise that this is an observational parallel: there is no suggestion that δ_S plays the role of a fundamental constant analogous to Planck's constant $\hbar$, or that quantum mechanics operates at planetary scales.

The recurrence of similar algebraic forms across different physical contexts has at times raised broader questions about underlying organising principles. The present work identifies such a pattern and makes it available for future theoretical investigation; developing any unified framework is beyond the empirical scope of this article.

8. Conclusions

The analysis presented in this work shows that a simple family of rational functions in the silver ratio $\delta_S = 1 + \sqrt{2}$ can reproduce the semi-major axes of Solar System planets with sub-per cent accuracy. Using the SRHF, we obtain a single algebraic sequence that:

- provides a compact description of planetary architecture from Mercury to Pluto;
- produces MAPE $\approx 0.72\%$ and RMSE ≈ 0.11 AU, improving on classical prescriptions by approximately an order of magnitude;
- naturally identifies a preferred node near 2.14 AU between Mars and Jupiter, acting as a harmonic boundary between inner and outer planets;
- satisfies a basic falsifiability criterion: dynamical or population-level signatures near the predicted *Harmonia* node, and silver-ratio ladders in exoplanetary systems, provide clear observational tests.

These results indicate that the Solar System's orbital configuration can be effectively summarised by a quantised harmonic model linked to $\sqrt{2}$ and δ_S. At this stage, this should be considered a phenomenological description rather than a complete planetary formation theory. Whether this pattern represents more fundamental disk dynamics, formation physics, or a compact encoding of complex processes remains an open question.

Future studies can examine: (i) resonant or density features near the *Harmonia* node; (ii) comparison with known exoplanetary systems; and (iii) exploration of how silver-ratio-like ladders might become stable attractors in disk evolution models. This study is intended as a hypothesis-generating step: isolating a mathematically simple pattern and making it available for future observational testing.

Computational Methods

All numerical evaluations, error metrics (MAPE, RMSE), and optimisation of the *Harmonia* position were computed with double-precision Python code (NumPy/SciPy). Model predictions used the fixed algebraic constants $A = \sqrt{2}$, $B = 1 + \sqrt{2}$, $C = 2\sqrt{2} - 1$, $D = \sqrt{2} - 1$. Observed semi-major axes (AU) were taken from JPL Horizons ephemerides (JPL Solar System Dynamics, 2023) and represent current best estimates (Murray and Dermott, 1999): Mercury 0.387, Venus 0.722, Earth 1.000, Mars 1.524, Jupiter 5.204, Saturn 9.559, Uranus 19.185, Neptune 30.156, Pluto 39.482 (Stern and New Horizons Team, 2015). Computational scripts used to generate all figures and residual analyses are provided in the associated GitHub and Zenodo archives.

Conflict of Interest

The author declares no commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

Salah-Eddin Gherbi conceived the harmonic framework, developed the mathematical model, implemented the numerical analysis, generated the figures, and wrote the manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Acknowledgements

The author acknowledges historical inspiration from Johannes Kepler and the wider tradition of harmonic and geometric approaches to cosmic structure, and thanks early readers for helpful comments on clarity and presentation. The geometric foundations of the Silver-Ratio Harmonic Framework, including the Celtic Cross construction and its relationship to the constants (A, B, C, D), are developed in fuller detail in the author's accompanying book (Gherbi, 2026).

Data Availability

The numerical datasets (planetary semi-major axes), Python scripts for computing the harmonic sequence, and all figure-generation notebooks are available in the project's GitHub repository:

`https://github.com/salahealer9/harmonic-architecture-solar-system`

A version-controlled archival copy is preserved on Zenodo under the associated project DOIs.

References

Salah-Eddin Gherbi. *Scala Harmonica: The Geometry of Planetary Resonance*. Vektor Publishing, 2026. ISBN 9781837095209.

JPL Solar System Dynamics. JPL horizons on-line ephemeris system, 2023. URL https://ssd.jpl.nasa.gov/horizons/. Accessed 2025-11-27.

Johannes Kepler. *Harmonices Mundi*. Johannes Planck, Linz, 1619.

Jacques Laskar. The chaotic motion of the solar system: A numerical estimate of the size of the chaotic zones. *Icarus*, 88(2):266–291, 1990. doi: 10.1016/0019-1035(90)90084-M.

C. D. Murray and S. F. Dermott. *Solar System Dynamics*. Cambridge University Press, Cambridge, 1999. ISBN 9780521575973. doi: 10.1017/CBO9781139174817.

Michael M. Nieto. *The Titius–Bode Law of Planetary Distances: Its History and Theory*. Pergamon Press, 1972.

S. N. Raymond and A. Morbidelli. Planetary migration in the context of planet formation. In H. J. Deeg and J. A. Belmonte, editors, *Handbook of Exoplanets*. Springer, Cham, 2018. doi: 10.1007/978-3-319-55333-7_167.

S. A. Stern and New Horizons Team. The pluto system: Initial results from the new horizons flyby. *Science*, 350(6258):1815, 2015. doi: 10.1126/science.aad1815.

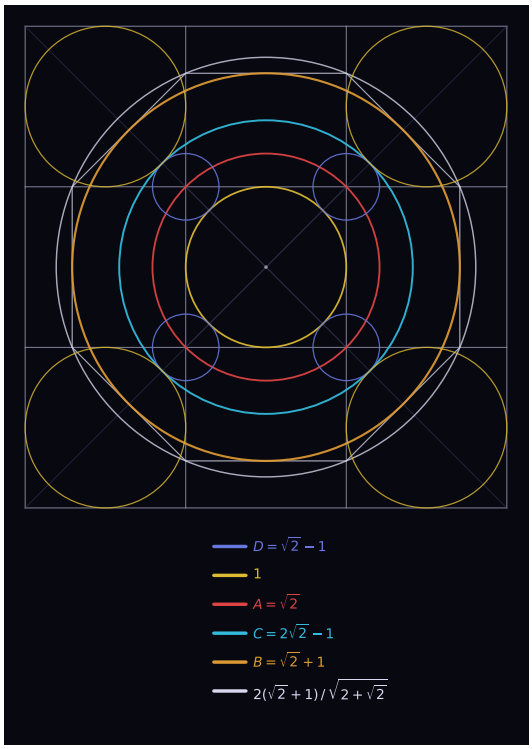


Figure 1. Geometric construction underlying the Silver-Ratio Harmonic Framework (SRHF). The 3×3 grid of equal unit squares, with concentric circles and four corner circles, generates the harmonic constants $A = \sqrt{2}$, $B = 1 + \sqrt{2}$, $C = 2\sqrt{2} - 1$, and $D = \sqrt{2} - 1$ directly from the geometry of the Celtic Cross. The outermost circle has circumradius $2(1 + \sqrt{2})/\sqrt{2 + \sqrt{2}}$, inscribed within a regular octagon whose inradius equals $B = \sqrt{2} + 1$, the Silver Ratio δ_S.

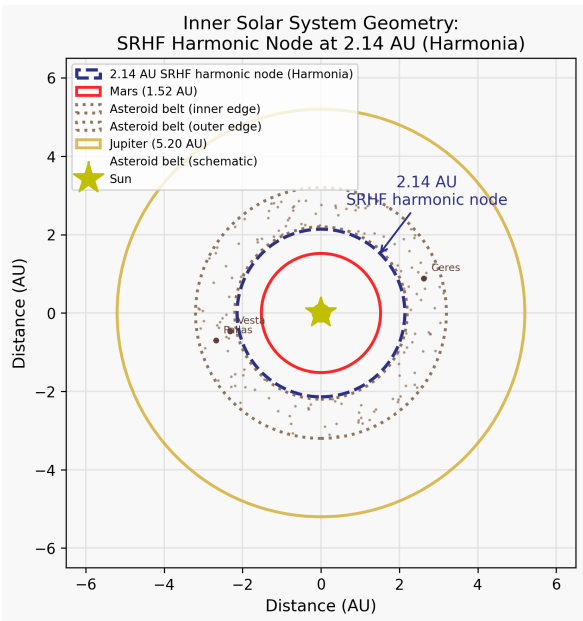
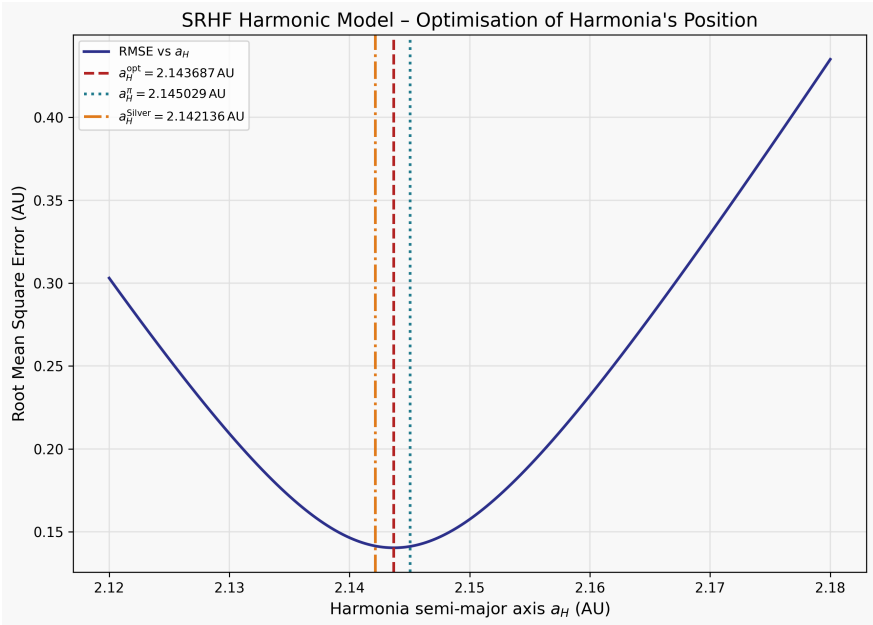
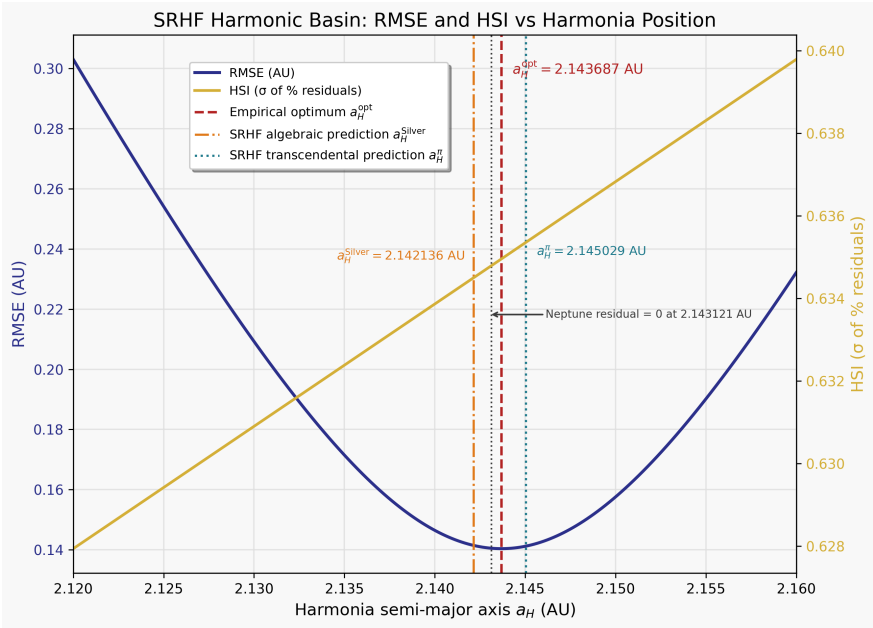


Figure 2. Schematic orbit at the 2.14 AU *Harmonia* node predicted by the SRHF. This location lies slightly inside the inner boundary of the main asteroid belt.

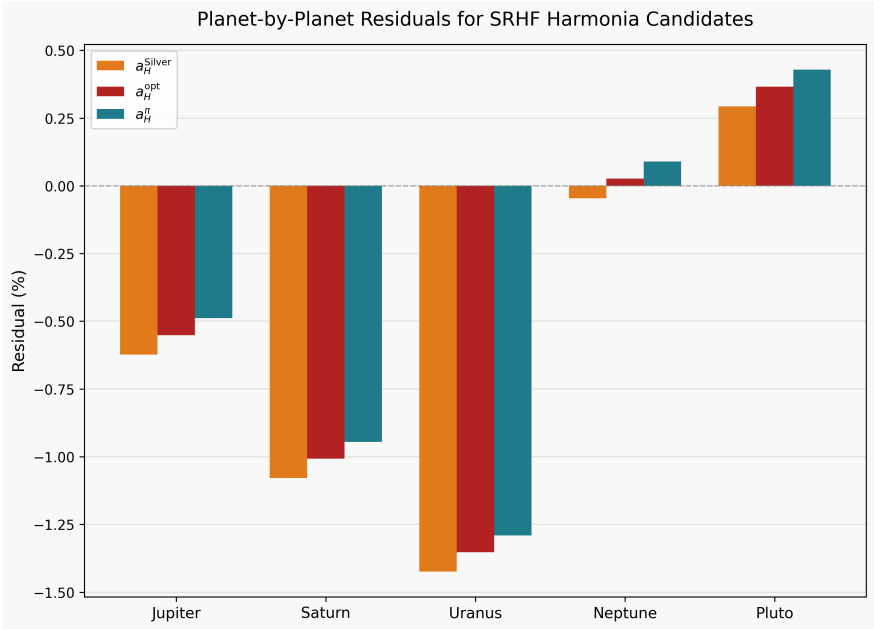


(a) Harmonia optimisation curve.

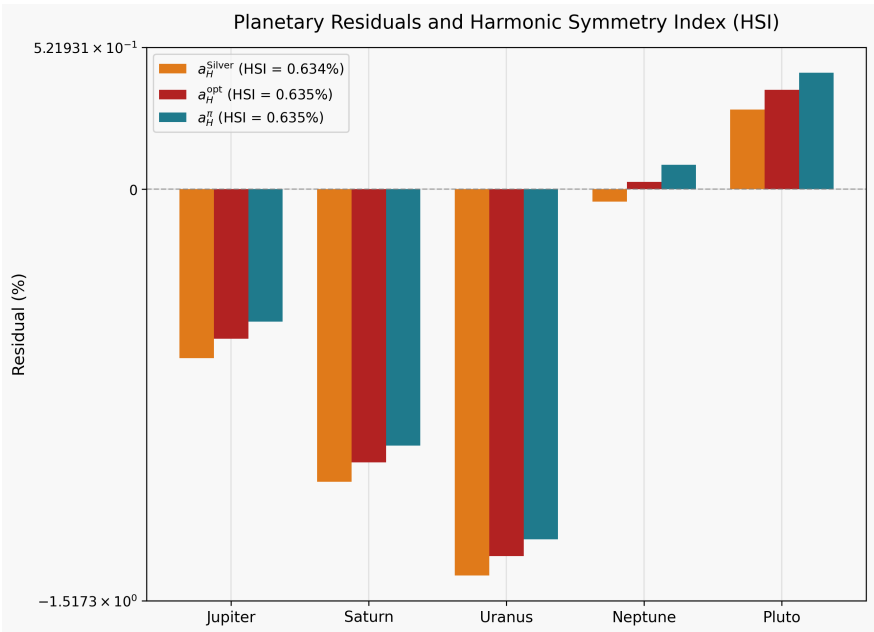


(b) Harmonic equilibrium basin.

Figure 3. Harmonia node optimisation and stability analysis. **(A)** RMSE optimisation showing minimum at $a_H^{\text{opt}} = 2.1437$ AU, coinciding with both the algebraic Silver Ratio prediction a_H^{Silver} and the transcendental solution $a_H^{\pi} = \pi^{2/3}$. **(B)** RMSE (indigo) and HSI (gold) showing extended stability plateau around the optimum.

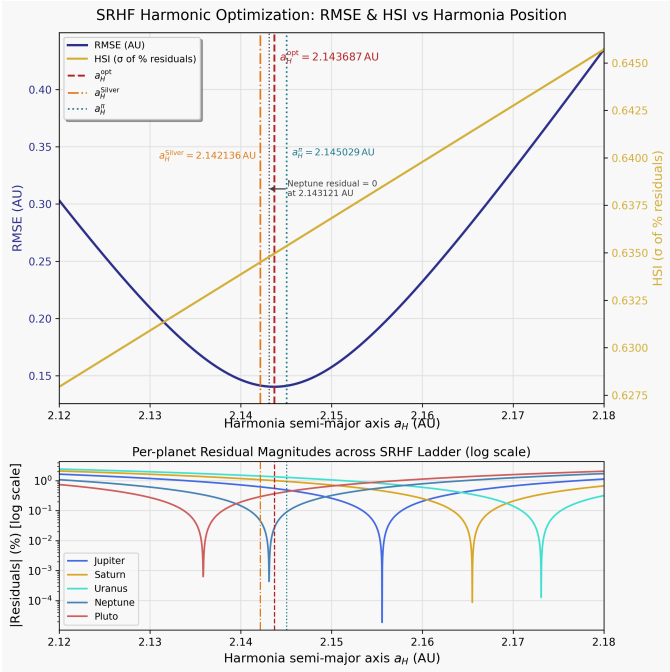


(a) Residuals comparison.

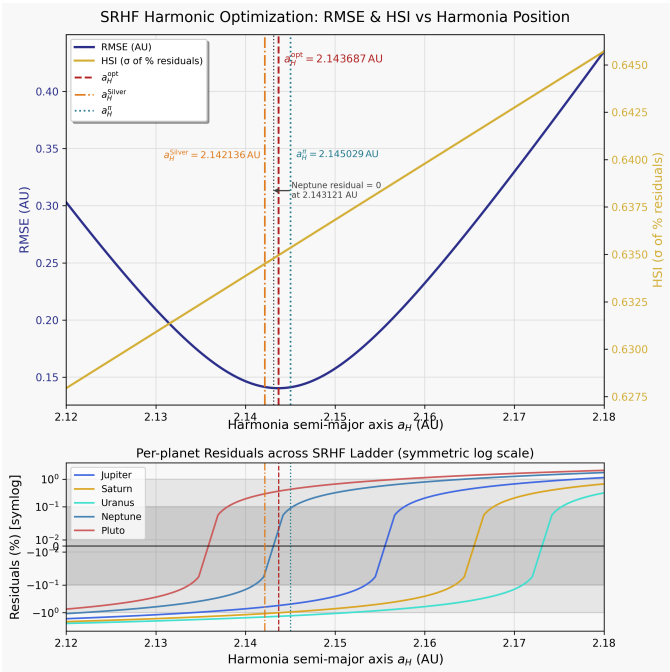


(b) Harmonic Symmetry Index.

Figure 4. Residual analysis and harmonic stability. **(A)** Percentage residuals for the algebraic Silver Ratio prediction a_H^{Silver}, empirically optimised a_H^{opt}, and a_H^{π} nodes. **(B)** HSI showing narrow stability basin centred on empirical optimum with HSI $\approx 0.635\%$.

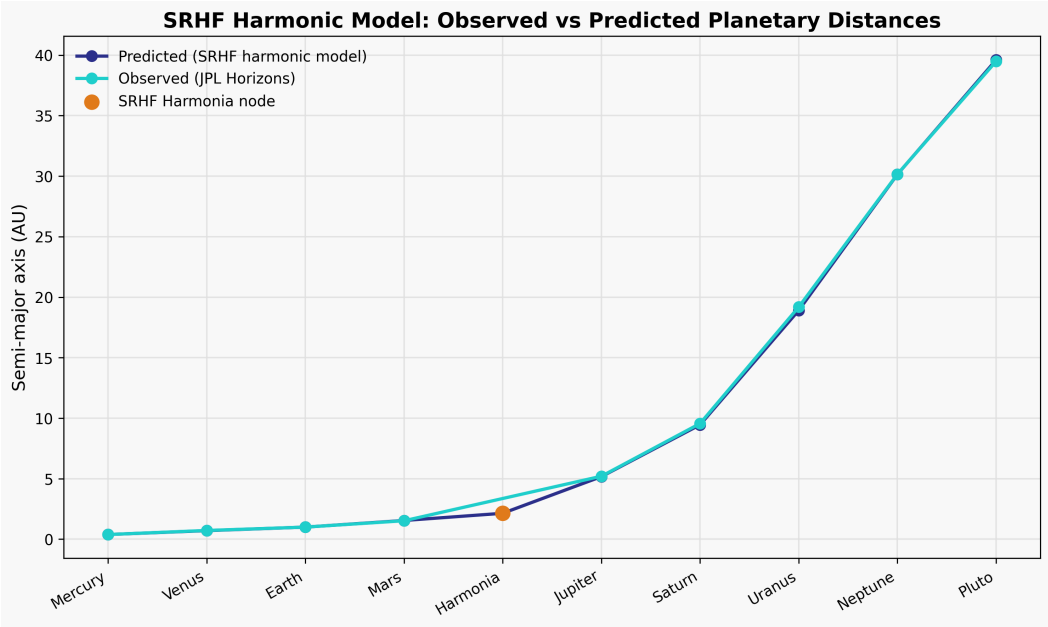


(a) Logarithmic residuals.

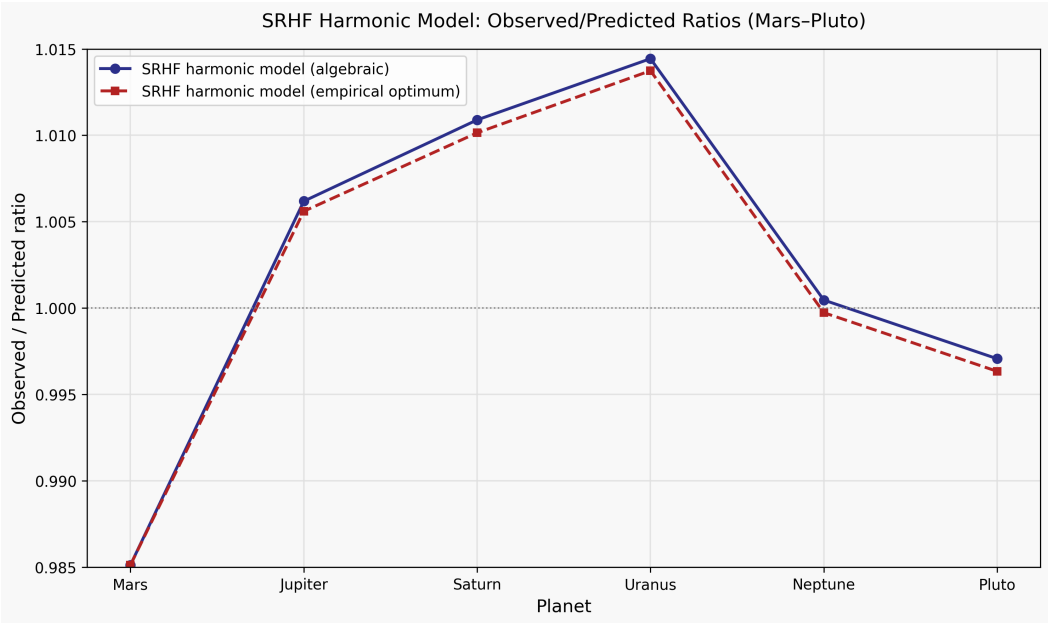


(b) Symlog residuals.

Figure 5. Logarithmic-scale residual analysis. **(A)** Logarithmic residuals showing Neptune's zero crossing at the empirical optimum. **(B)** Symmetric logarithmic residuals revealing harmonic inversion and the stability region.

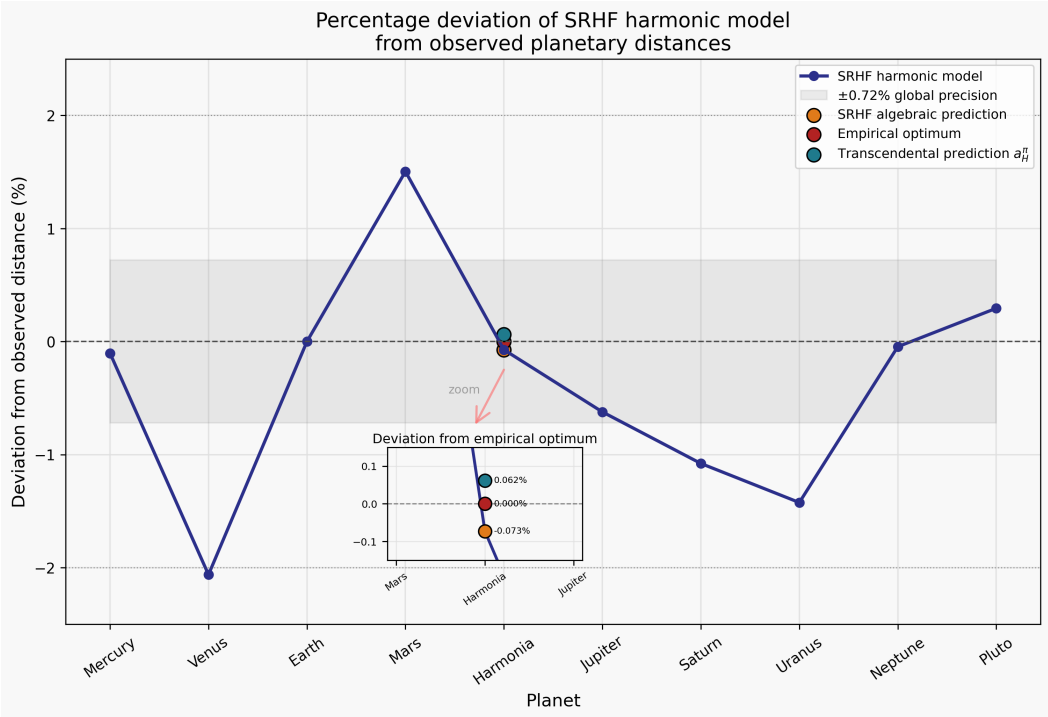


(a) Full-system ratios.

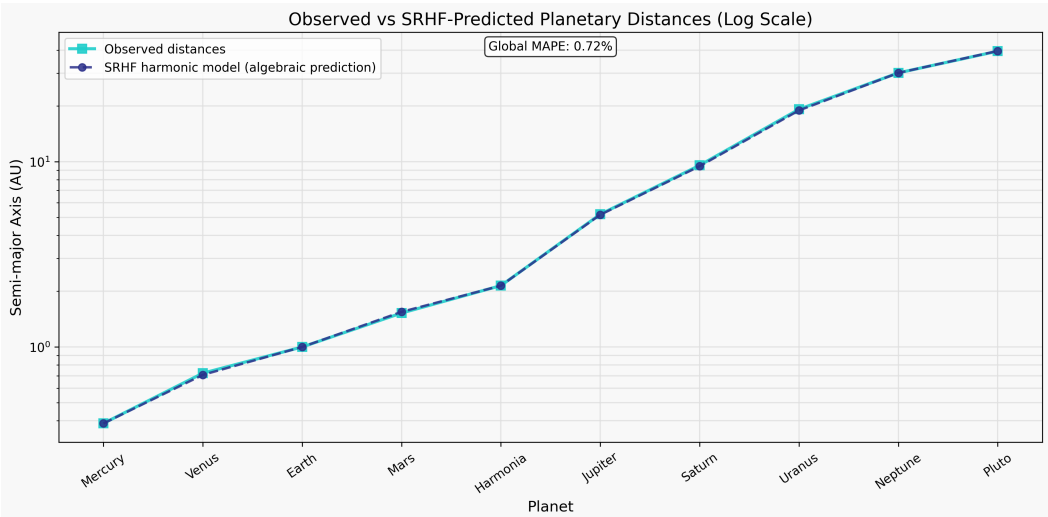


(b) Mars–Pluto zoom.

Figure 6. Observed-to-predicted ratios for planetary semi-major axes. **(A)** Full-system ratios from Mercury to Pluto, showing that modelled radii remain close to unity. **(B)** Zoom on the Mars–Pluto range highlighting $\lesssim 1.5\%$ deviations and the position of the Harmonia node near 2.14 AU.



(a) Percentage deviations.



(b) Log-scale validation.

Figure 7. Model validation across scales. **(A)** Percentage deviations with 0.72% mean absolute error band. **(B)** Observed and predicted semi-major axes on a logarithmic scale, maintaining MAPE $\approx$ 0.72% across the full system.

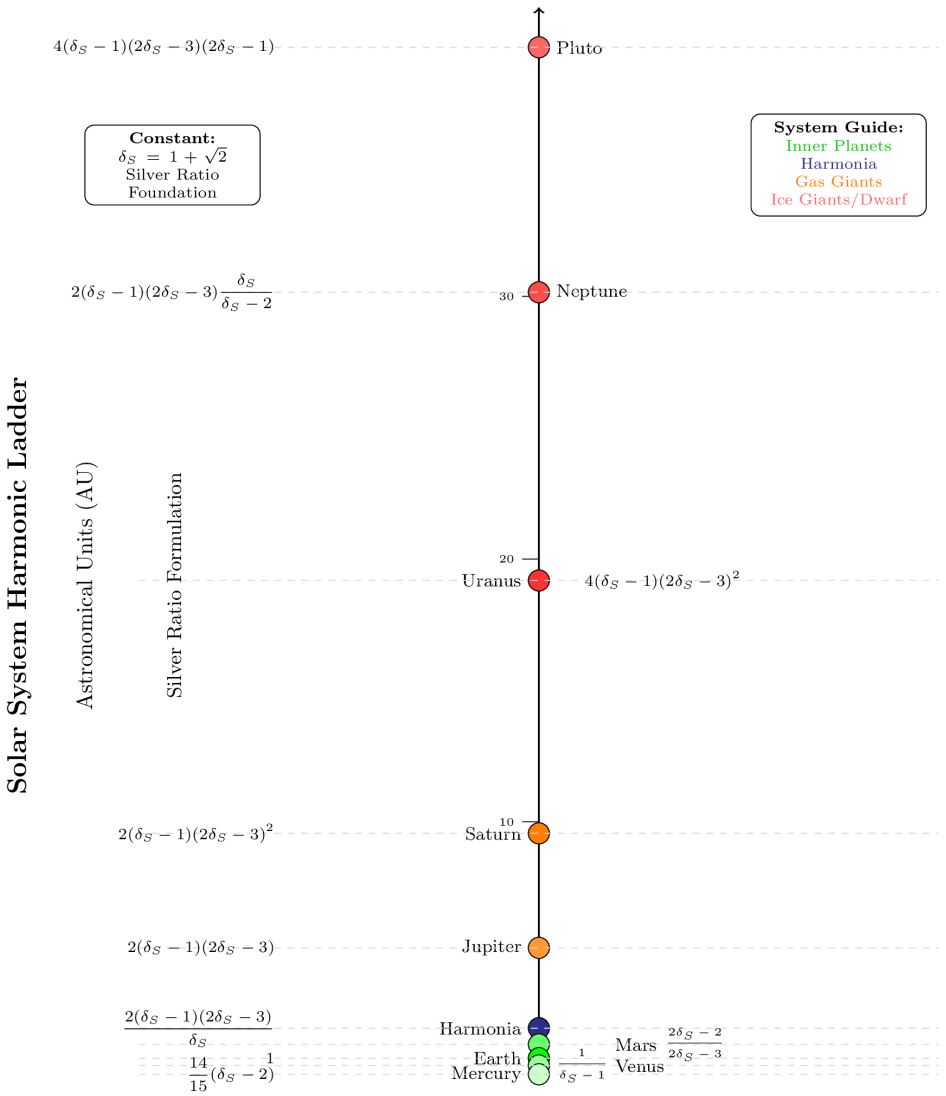


Figure 8. The complete Solar System harmonic ladder expressed in terms of the Silver Ratio $\delta_S = 1 + \sqrt{2}$. Each planet occupies a quantised level defined by rational functions of δ_S, demonstrating that planetary spacing follows a precise algebraic hierarchy.

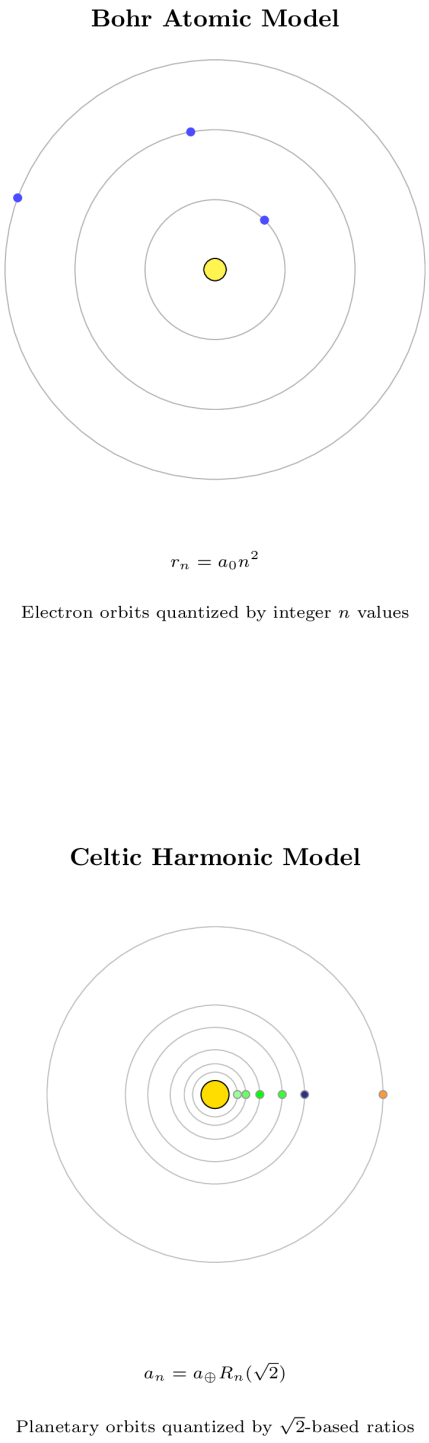


Figure 9. Quantum–Celestial Analogy: Bohr Atomic Model vs. SRHF model. A vertical comparison highlights the structural similarity of discrete levels in both systems: atomic orbits from quantum numbers, planetary orbits from algebraic harmonics of $\sqrt{2}$ in the phenomenological model.

Table 1. SRHF model vs. observed semi-major axes.

Planet	**Observed (AU)**	**Model (AU)**	**Δ (AU)**	**Dev. (%)**
Mercury	0.387	0.387	−0.000	−0.10
Venus	0.722	0.707	−0.015	−2.06
Earth	1.000	1.000	±0.000	0.00
Mars	1.524	1.547	+0.023	+1.48
Harmonia	2.144	2.142	−0.002	−0.07
Jupiter	5.204	5.172	−0.032	−0.62
Saturn	9.559	9.456	−0.103	−1.08
Uranus	19.185	18.912	−0.273	−1.42
Neptune	30.156	30.142	−0.014	−0.05
Pluto	39.482	39.598	+0.116	+0.29
MAPE				**0.72%**
RMSE			**0.11 AU**	

Table 2. Comparative performance of historical models and the SRHF model.

Metric	**Kepler (1619)**	**Titius–Bode (1772)**	**SRHF (2026)**
Functional basis	Platonic solids	Empirical sequence	Algebraic $\sqrt{2}$ ratios
MAPE	10.3%	2.5%	**0.72%**
RMSE (AU)	0.24	0.22	**0.11**
Predictive scope	6 planets	8 planets (fails outer)	**9 + Harmonia**
Underlying principle	Nested geometry	Empirical pattern	**Algebraic quantisation**

Table 3. Illustrative structural analogy between selected concepts in atomic physics and in the SRHF parameterisation of planetary spacing. The correspondence is formal and phenomenological only, and is not used in the derivation or quantitative assessment of the model.

Quantum Systems (Atomic Scale)	**SRHF (Planetary Scale)**
Discrete electron orbitals	Discrete harmonic orbital radii in the parameterisation
Quantum numbers (n, l, m)	Integer indices labelling levels of the δ_S ladder
Energy-level spacings	Spacing of modelled semi-major axes from rational functions of δ_S
Radial wavefunction structure	Radial pattern of modelled orbital radii and resonances
Stationary bound states	Long-lived orbital configurations (phenomenological)

Note: The analogies listed above are structural rather than physical. They are intended to assist conceptual interpretation and do not imply quantum effects or microscopic mechanisms operating at planetary scales.